

University of Dhaka
Department of Computer Science and Engineering
Professional Masters in Information and Cyber Security (PMICS)

PROJECT PROPOSAL FORM for CSE-810: Project on Cyber Security

1.	Proposed title of the project:	
	An Autonomous Multi-Agent Red Team for Intelligent Vulnerability Assessment using Agentic RAG and Human-in-the-Loop Safety Controls	
2.	Name of student 1: Nishan Paul Registration number: 55 Contact number: 0184308377	Batch: 05
	Name of student 2: Md Rakibur Rahman Registration number: 49 Contact number: 01730317736	Batch: 05
3.	Brief description of the proposed project (Importance of the project/research domain, motivation, goals, and expected outcomes – Not more than 200 words) The increasing sophistication of cyber threats demands equally intelligent and automated security assessment tools. Existing penetration testing approaches rely heavily on manual effort, are time-consuming, and fail to scale with the complexity of modern IT infrastructures. CMatrix addresses this gap by implementing an autonomous multi-agent security orchestration platform powered by Large Language Models (LLMs) and the LangGraph framework. The system employs a hierarchical Master-Worker agent architecture, where a central Supervisor Agent coordinates seven specialized worker agents—covering network scanning, web application security, authentication testing, configuration auditing, API security, vulnerability intelligence, and command execution. Each agent operates as an independent reasoning unit with its own LLM instance and domain-specific toolset. A key research contribution is the integration of Agentic Retrieval-Augmented Generation (RAG) for CVE intelligence, featuring query reformulation, multi-hop CVE graph traversal, semantic reranking, and self-correcting search loops. The platform also implements a Human-in-the-Loop (HITL) approval framework with risk-level gating, ensuring dangerous operations require explicit human authorization before execution. The expected outcomes include a production-grade, deployable security assessment platform with a web-based interface, real-time streaming, LangGraph state checkpointing, Qdrant vector memory, and comprehensive audit logging—capable of autonomously performing security assessments that rival the output of senior security analysts.	

4. **Methodology (Tools, methods, platforms, and techniques that will be used to carry out the project/research works – Not more than 150 words)**

Architecture: LangGraph-based multi-agent system using the Supervisor pattern with subgraph delegation. Each specialized agent implements the ReAct (Reasoning + Acting) loop with up to five autonomous reasoning iterations.

Advanced Reasoning: ReWOO (Reasoning Without Observation) for upfront scan planning, Self-Reflection for autonomous quality assessment, and Tree of Thoughts for multi-strategy evaluation and selection.

Agentic RAG Pipeline: Query reformulation engine (LLM-based synonym expansion and CPE extraction), multi-hop CVE graph traversal (NetworkX, depth-limited to 3 hops), BGE-reranker for semantic reranking, self-correcting search loops, and a local CVE vector store (Qdrant) seeded from the NIST NVD feed with 50,000+ CVEs.

Technology Stack: Python 3.12, FastAPI, LangChain, LangGraph, PostgreSQL (state + checkpointing), Redis (job queue + semantic caching), Qdrant (vector database), Celery (background jobs), Next.js (frontend with SSE streaming).

Evaluation: Unit testing (>90% coverage), integration testing against a deliberately vulnerable lab environment, and benchmarking of planning efficiency, CVE retrieval accuracy, and HITL responsiveness.

5. **You may mention at most two names for the expected supervisor of the project**

i) Dr. Md. Abdur Razzaque

ii)



6. **You may mention at most two names for the expected co-supervisor of the project**

i)

ii)

N.B.: A project will be guided by a supervisor and a co-supervisor, if necessary. Please note that at least one of them will be from the University of Dhaka, as determined by the concerned examination committee of the semester.

7. Nishan Paul 03/04/2026
Signature of student 1 with date

MRAR 03-04-26
Signature of student 2 with date

8. (To be filled by the chair of the concerned examination committee)

Name of the assigned supervisor:

Name of the assigned co-supervisor:

(Signature of Chair of Examination Committee with date)